

MongoDB_Querying_Code

March 21, 2026

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[ ]: #Querying code by Rudy

[ ]: #Import Necessary libraries
import matplotlib.pyplot as plt
import pymongo

[ ]: #Get collection
MONGO_URI = "mongodb://localhost:27017/?
↳directConnection=true&serverSelectionTimeoutMS=2000"
DB_NAME = 'sf_restaurants_project'
COLLECTION_NAME = "restaurants_joined"
client = pymongo.MongoClient(MONGO_URI)
db = client[DB_NAME]
collection = db[COLLECTION_NAME]

[ ]: #Query to get average lifespans for restaurants on/not on The Infatuation
lifespan_by_infatuation = collection.aggregate([
    {'$match': {}},
    {'$group': {'_id': "$match_found", 'total': {'$avg': "$lifespan_days"}}}]
])
lifespan_by_infatuation = list(lifespan_by_infatuation)

[ ]: #Put results into lists
featured = []
lifespans = []
for row in lifespan_by_infatuation:
    if row['_id'] == True:
        featured.append('Featured')
    else:
        featured.append('Not Featured')
for row in lifespan_by_infatuation:
    lifespans.append(row["total"])

[ ]: #Print lifespans for featured/unfeatured restaurants
print("Average Lifespans for Restaurants That Were or Were not Featured on_
↳Infatuation")
for feature, lifespan in zip(featured, lifespans):
    print(f"{feature}: {lifespan:.0f} days")
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Average Lifespans for Restaurants That Were or Were not Featured on Infatuation
Not Featured: 2092 days
Featured: 3047 days

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[ ]: #Query to get 5 neighborhoods with the best lifespans
best_lifespan_neighborhoods = collection.aggregate([
    {'$match': {}},
    {'$group': {'_id': "$sf_neighborhood", 'total': {'$avg': ␣
↪"$lifespan_days"}}},
    {'$sort': {'total': -1}},
    {'$limit': 5}
])
best_lifespan_neighborhoods = list(best_lifespan_neighborhoods)
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[ ]: #Query to get 5 neighborhoods with the worst lifespans
worst_lifespan_neighborhoods = collection.aggregate([
    {'$match': {}},
    {'$group': {'_id': "$sf_neighborhood", 'total': {'$avg': ␣
↪"$lifespan_days"}}},
    {'$sort': {'total': 1}},
    {'$limit': 5}
])
worst_lifespan_neighborhoods = list(worst_lifespan_neighborhoods)
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[ ]: #Print 5 Neighborhoods With The Best Lifespans
print('Neighborhoods With The Best Lifespans: ')
for row in best_lifespan_neighborhoods:
    print(f"{row['_id']}: {row['total']:.0f} days")
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Neighborhoods With The Best Lifespans:
Lincoln Park: 6268 days
Chinatown: 3233 days
Glen Park: 2953 days
Presidio Heights: 2852 days
Haight Ashbury: 2756 days

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[ ]: #Print 5 Neighborhoods With The Worst Lifespans
print('Neighborhoods With The Worst Lifespans: ')
for row in worst_lifespan_neighborhoods:
    print(f"{row['_id']}: {row['total']:.0f} days")
```

Neighborhoods With The Worst Lifespans:
Seacliff: 480 days
Treasure Island: 688 days
Visitacion Valley: 1008 days
Twin Peaks: 1092 days
Oceanview/Merced/Ingleside: 1154 days

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[ ]: #Get ratings and lifespans for all restaurants we have
lifespans_and_ratings = collection.find({'rating': {'$ne': None}}, {'lifespan_days':1, 'rating':1})
lifespans_and_ratings = list(lifespans_and_ratings)
scatter_lifespans = [row['lifespan_days'] for row in lifespans_and_ratings]
scatter_ratings = [row['rating'] for row in lifespans_and_ratings]

[ ]: #Create scatter plot of ratings and lifespans
plt.scatter(scatter_ratings, scatter_lifespans)
plt.title("Lifespan vs. Infatuation Rating")
plt.xlabel("Infatuation Rating")
plt.ylabel("Lifespan(days)")

[ ]: Text(0, 0.5, 'Lifespan(days)')
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